arXiv:submit/4812508 [cs.CL] 27 Mar 2023

# GPT-4 Technical Report

**OpenAI***

## Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

## 1 Introduction

This technical report presents GPT-4, a large multimodal model capable of processing image and text inputs and producing text outputs. Such models are an important area of study as they have the potential to be used in a wide range of applications, such as dialogue systems, text summarization, and machine translation. As such, they have been the subject of substantial interest and progress in recent years [1–34].

One of the main goals of developing such models is to improve their ability to understand and generate natural language text, particularly in more complex and nuanced scenarios. To test its capabilities in such scenarios, GPT-4 was evaluated on a variety of exams originally designed for humans. In these evaluations it performs quite well and often outscores the vast majority of human test takers. For example, on a simulated bar exam, GPT-4 achieves a score that falls in the top 10% of test takers. This contrasts with GPT-3.5, which scores in the bottom 10%.

On a suite of traditional NLP benchmarks, GPT-4 outperforms both previous large language models and most state-of-the-art systems (which often have benchmark-specific training or hand-engineering). On the MMLU benchmark [35, 36], an English-language suite of multiple-choice questions covering 57 subjects, GPT-4 not only outperforms existing models by a considerable margin in English, but also demonstrates strong performance in other languages. On translated variants of MMLU, GPT-4 surpasses the English-language state-of-the-art in 24 of 26 languages considered. We discuss these model capability results, as well as model safety improvements and results, in more detail in later sections.

This report also discusses a key challenge of the project, developing deep learning infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to make predictions about the expected performance of GPT-4 (based on small runs trained in similar ways) that were tested against the final run to increase confidence in our training.

Despite its capabilities, GPT-4 has similar limitations to earlier GPT models [1, 37, 38]: it is not fully reliable (e.g. can suffer from "hallucinations"), has a limited context window, and does not learn

*Please cite this work as "OpenAI (2023)". Full authorship contribution statements appear at the end of the document. Correspondence regarding this technical report can be sent to gpt4-report@openai.com

from experience. Care should be taken when using the outputs of GPT-4, particularly in contexts where reliability is important.

GPT-4's capabilities and limitations create significant and novel safety challenges, and we believe careful study of these challenges is an important area of research given the potential societal impact. This report includes an extensive system card (after the Appendix) describing some of the risks we foresee around bias, disinformation, over-reliance, privacy, cybersecurity, proliferation, and more. It also describes interventions we made to mitigate potential harms from the deployment of GPT-4, including adversarial testing with domain experts, and a model-assisted safety pipeline.

## 2 Scope and Limitations of this Technical Report

This report focuses on the capabilities, limitations, and safety properties of GPT-4. GPT-4 is a Transformer-style model [39] pre-trained to predict the next token in a document, using both publicly available data (such as internet data) and data licensed from third-party providers. The model was then fine-tuned using Reinforcement Learning from Human Feedback (RLHF) [40]. Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar.

We are committed to independent auditing of our technologies, and shared some initial steps and ideas in this area in the system card accompanying this release.[2] We plan to make further technical details available to additional third parties who can advise us on how to weigh the competitive and safety considerations above against the scientific value of further transparency.

## 3 Predictable Scaling

A large focus of the GPT-4 project was building a deep learning stack that scales predictably. The primary reason is that for very large training runs like GPT-4, it is not feasible to do extensive model-specific tuning. To address this, we developed infrastructure and optimization methods that have very predictable behavior across multiple scales. These improvements allowed us to reliably predict some aspects of the performance of GPT-4 from smaller models trained using $1,000\times - 10,000\times$ less compute.

### 3.1 Loss Prediction

The final loss of properly-trained large language models is thought to be well approximated by power laws in the amount of compute used to train the model [41, 42, 2, 14, 15].

To verify the scalability of our optimization infrastructure, we predicted GPT-4's final loss on our internal codebase (not part of the training set) by fitting a scaling law with an irreducible loss term (as in Henighan et al. [15]): $L(C) = aC^b + c$, from models trained using the same methodology but using at most 10,000x less compute than GPT-4. This prediction was made shortly after the run started, without use of any partial results. The fitted scaling law predicted GPT-4's final loss with high accuracy (Figure 1).

### 3.2 Scaling of Capabilities on HumanEval

Having a sense of the capabilities of a model before training can improve decisions around alignment, safety, and deployment. In addition to predicting final loss, we developed methodology to predict more interpretable metrics of capability. One such metric is pass rate on the HumanEval dataset [43], which measures the ability to synthesize Python functions of varying complexity. We successfully predicted the pass rate on a subset of the HumanEval dataset by extrapolating from models trained with at most $1,000\times$ less compute (Figure 2).

For an individual problem in HumanEval, performance may occasionally worsen with scale. Despite these challenges, we find an approximate power law relationship $-\mathrm{E}_P[\log(\mathrm{pass_rate(C)})] = \alpha * \mathrm{C}^{-k}$

[2] In addition to the accompanying system card, OpenAI will soon publish additional thoughts on the social and economic implications of AI systems, including the need for effective regulation.

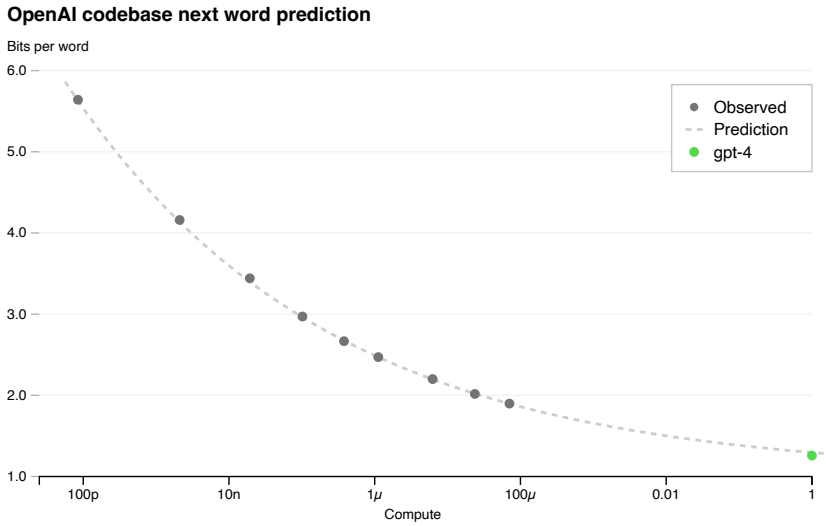



**Figure 1.** Performance of GPT-4 and smaller models. The metric is final loss on a dataset derived from our internal codebase. This is a convenient, large dataset of code tokens which is not contained in the training set. We chose to look at loss because it tends to be less noisy than other measures across different amounts of training compute. A power law fit to the smaller models (excluding GPT-4) is shown as the dotted line; this fit accurately predicts GPT-4's final loss. The x-axis is training compute normalized so that GPT-4 is 1.

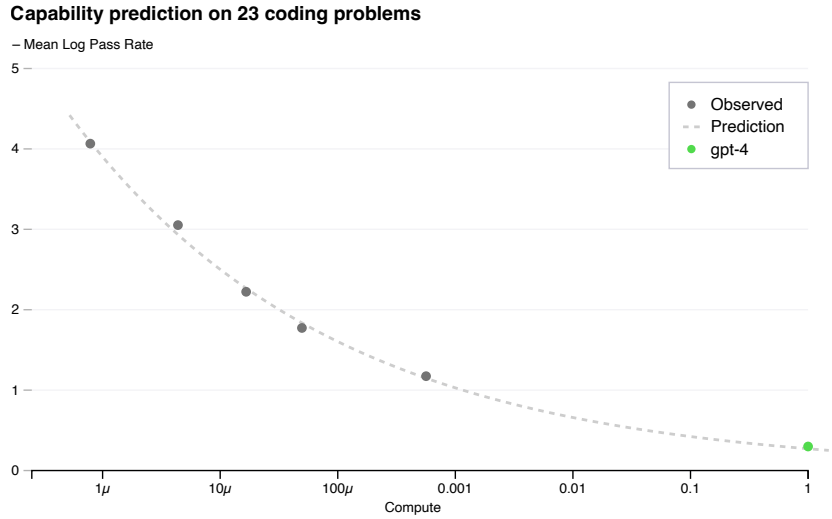



**Figure 2.** Performance of GPT-4 and smaller models. The metric is mean log pass rate on a subset of the HumanEval dataset. A power law fit to the smaller models (excluding GPT-4) is shown as the dotted line; this fit accurately predicts GPT-4's performance. The x-axis is training compute normalized so that GPT-4 is 1.